Study Guide for CMIT 265

uCertify CompTIA Network+ (N10-008)

Week	Topics	Exam Objectives
1	**1. The Open Systems Interconnection Specifications** • The OSI Reference Model • The OSI Model • TCP/IP Stack **2. Network Topologies and Types** • Defining a Network • Network types and characteristics • Networks defined based on resource location • Networks defined by topology • Virtual Networks • Provider links **3. Network Media Types** • Copper and Fiber Media and Connectors • Unshielded Twisted Pair • Plenum vs. Nonplenim cable • Multiplexing in Fiber-Optic Networks • Cable management	1.1, 1.2, 1.3
2	**5. Common ports and protocols** • Ports and Protocols • IP Protocol types **6. Network Services** • DHCP • DNS • NTP **7. Corporate and Datacenter Architectures** • The three-tiered network architecture • Software-Defined networking • Storage area networking • Deciding on an architecture **8. Cloud Concepts** • Deployment models • Service Models • Key Cloud concepts **9. Various Network Devices** • Networking Devices • Networked Devices	1.5, 1.6, 1.7, 2.1

3	**4. IP Addressing** • Binary numbering • IPv4 Addressing • Assigning IPv4 Addresses • Subnetting • Address Translation • IPv6	1.4
4	**10. Routing Technologies and Bandwidth Management** • Routing • Sources of Routing Information • Routing Protocol Characteristics • Bandwidth Management **11. Ethernet Switching** • Principles of Ethernet • Ethernet Switch Features	2.2, 2.3
5	**12. Wireless Standards** • Introducing Wireless LANs • Deploying Wireless LANs • Securing Wireless LANs **13. Ensure Network Available** • Monitoring Tools • Additional Monitoring Topics **14. Organizational Documents and Polices** • Plans and Policies • Hardening and Security Policies **15. High Availability and Disaster Recovery** • High Availability	2.4, 3.1, 3.2, 3.3
6	**16. Security Concepts** • Core Security Concepts • Authentication Methods • Risk Management and SIEM **17. Types of Attacks** • Technology-Based Attacks • Human and Environmental Attacks **18. Network Hardening Techniques** • Best Practices • Wireless Security and IoT Considerations **19. Remote Access Methods** • Virtual Private Network (VPN) • Other Remote Access Technologies • Authentication and Authorization Considerations • In-Band vs. Out-of-Bank Management **20. Physical Security** • Detection Methods	4.1, 4.2, 4.3, 4.4, 4.5

	• Prevention Methods • Asset Disposal	
7	**21. A Network Troubleshooting Methodology** • Troubleshooting basics **22. Troubleshoot common cable problems** • Specifications and Limitations • Cable Considerations and Applications **23. Network software tools and commands** • Software Tools • Command Line Tools • Basic Network Platform Commands **24. Troubleshoot common wireless issues** • Wireless networking troubleshooting **25. Troubleshoot general network issues** • Considerations for general network troubleshooting	5.1, 5.2, 5.3, 5.4, 5.5